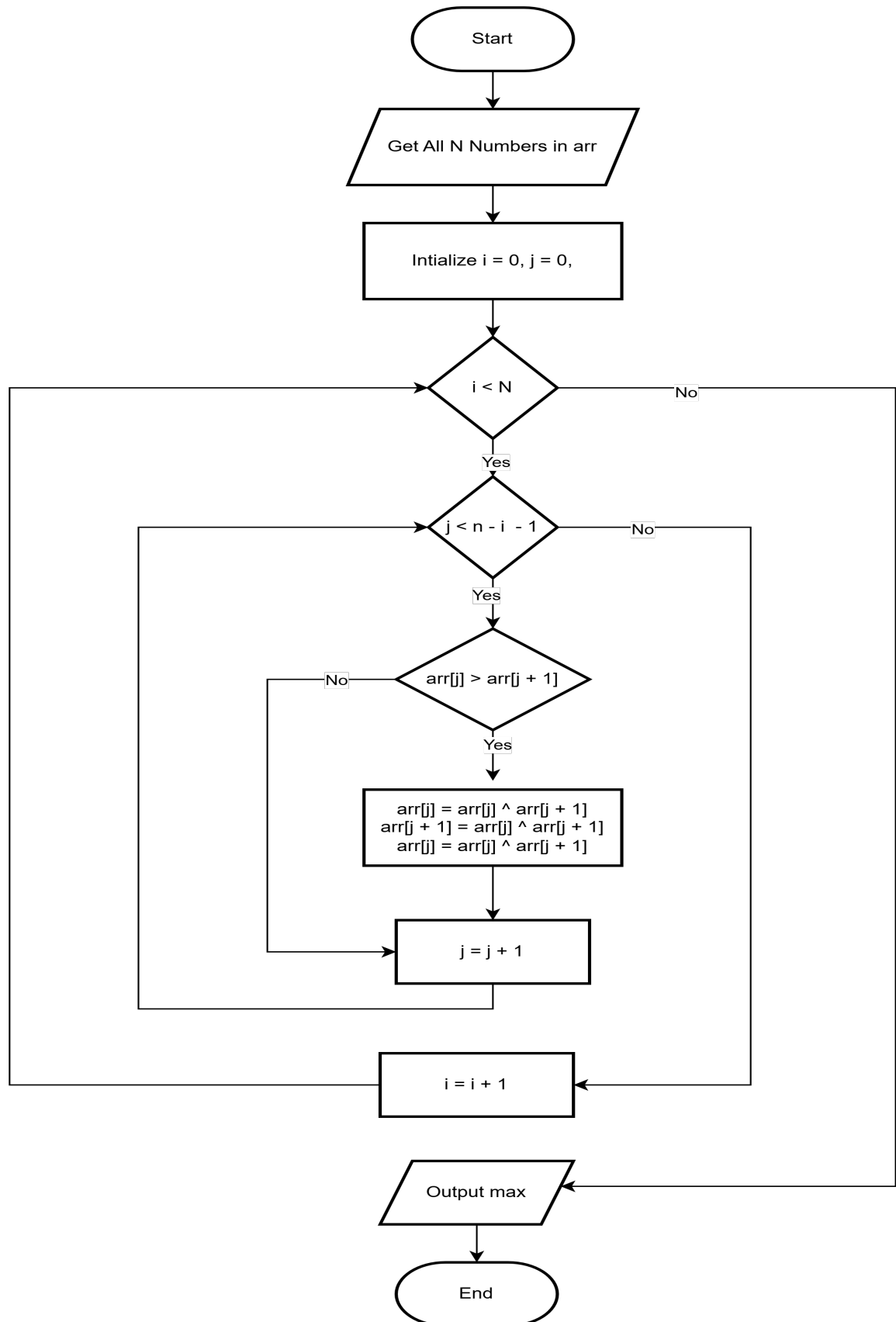


Assignment 1:

1. Draw a flow chart for bubble sort mechanism



2. Using bit manipulation, Swap the last two bits of a number.

```
num1 = int(input())
num2 = int(input())

# Extracts the last 2 bits.
num1_last_2bits = num1 & 3
num2_last_2bits = num2 & 3

# XOR the last 2 bits of the numbers to make it 0's.
num1 ^= num1_last_2bits
num2 ^= num2_last_2bits

# OR the number with that extracted bits to set those bits in that
number's last two bits.
num1 |= num2_last_2bits
num2 |= num1_last_2bits

print(num1, num2)
```

Assignment 2(coding)

1. Write a program to check if two strings are anagrams without using built-in functions.

```
def isAnagram(str1, str2):
    map = dict()
    # Checking for equal length
    if len(str1) != len(str2):
        return False

    n = len(str1)
    for i in range(n):
        # Counting the characters
        # Increment for First String
        map[str1[i]] = map.get(str1[i], 0) + 1
        # Decrement for the Second String
        map[str2[i]] = map.get(str2[i], 0) - 1
```

```
    s = set(map.values())
    if len(s) > 1:
        return False

    # checking the map only contains 0
    val = s.pop()
    if val == 0:
        return True
    else:
        return False

str1 = input()
str2 = input()
```

```
print(isAnagram(str1=str1, str2=str2))
```

2. Write a program to check if two strings are isomorphic.

```
def isIsomorphic(str1, str2):
    # For storing beginning indices of every characters
    map1 = dict()
    map2 = dict()

    for i in range(len(str1)):
        # storing the beginning index value of every first occurring
        # characters in respective maps
        if map1.get(str1[i], None) is None:
            map1[str1[i]] = i
        if map2.get(str2[i], None) is None:
            map2[str2[i]] = i

    # If Index mismatches then it is not in isomorphic
    if map2[str2[i]] != map1[str1[i]]:
        return False

    return True

str1 = input()
str2 = input()
print(isIsomorphic(str1, str2))
```